

1 Composition - AIP

Definition (source: <http://de.wikipedia.org/wiki/Gattierung>)

In foundry, composition (optimization of composition) means the make-up of foundry material that is melted down in the melting furnace. Composition is necessary to achieve an as exact chemical make-up of the cast material as possible without having to add many ingredients. The result is kept in a composition list acting as a bill of material.

However, it is required to have detailed knowledge about the chemical analysis of the raw materials, i.e. normally pig iron, steel scrap and recycled material. Based on this analysis, it is calculated in which proportion they have to be put into the furnace to achieve the required characteristics of the material.

Composition has to be optimized if the charge make-up is to be performed in a cost-effective manner. To do so, the required quantities of the raw materials, such as steel scrap or pig iron are calculated using mathematical procedures in order to achieve the required characteristics of the material by combining the raw materials as economically as possible. Therefore, an exact chemical analysis of raw materials is required. The mathematical calculation is performed using specialized computer software for the optimization of composition.

1.1 Perform charging

The dialog "perform charging" shows the current list of materials the furnace is fed with. The list includes the planned, reserved and already fed quantities as well as the required remaining quantities.

The quantities can be edited here.

Perform charging

Workplace 22222222 Casting

Order 2222222205 Operation 0100

Material	Materialdesc	Target qty	Unit	Reserved qt	Loaded qt	Materialbuffer	Remaining qt	Input buffer
8260	8260 // Al 97,95 Si 0,1	9000,000	kg	9000,000	9000,000	22222222E		9000,000
Al	aluminum (pure)	100,000	kg	100,000	50,000	22222222E		50,000
			kg			2222-03	50,000	0,000
Fe	iron (pure)	20,000	kg	20,000	15,000	22222222E		15,000
			kg			2222-01	5,000	0,000

Material: 8260

Quantity: KG

Material buffer: 22222222E

Staff badge number:

mpcv CTAIP V2.0.2.48 Server: win2008-6.1 TNR:222 22.01.13 09:13:08

Figure: Perform charging – C_CHPF

Data from the selected row are entered in the below input fields

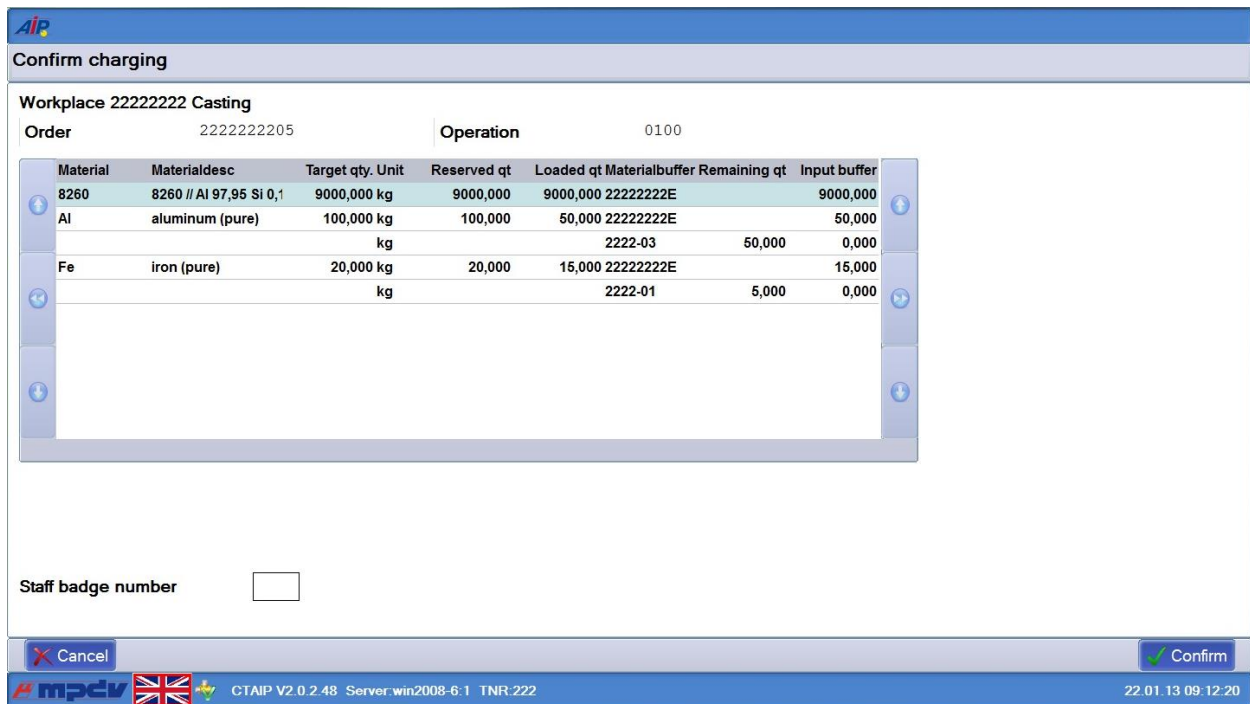
The button "post" prompts posting of the material in the machine's input buffer. Then the list is reloaded. For this purpose, the terminal needs to be connected online with the HYDRA server.

The dialog remains opened until the button "cancel" is clicked.

The list does not show the material that has already been confirmed by the dialog "confirm charging". Consequently, the list of the dialog "perform charging" is empty directly after finishing the charging process. Unplanned material may also be added to the dialog "perform charging". This material is then shown in the list.

1.2 Confirm charging

The current list of materials is shown again to confirm charging. Besides the badge number, no further input is required.



Confirm charging

Workplace 22222222 Casting

Order 2222222205 Operation 0100

Material	Materialdesc	Target qty	Unit	Reserved qt	Loaded qt	Materialbuffer	Remaining qt	Input buffer
8260	8260 // Al 97,95 Si 0,1	9000,000	kg	9000,000	9000,000	22222222E	9000,000	9000,000
Al	aluminum (pure)	100,000	kg	100,000	50,000	22222222E	50,000	50,000
						2222-03	50,000	0,000
Fe	iron (pure)	20,000	kg	20,000	15,000	22222222E	15,000	15,000
						2222-01	5,000	0,000

Staff badge number

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Figure: Confirm charging – C_CHCF

Once confirmed, the materials from the input buffer are posted onto the machine's output buffer. The reservation of remaining quantities of materials reserved for the charging order is cancelled.

1.4 Take sample

A sample is taken to check the actual composition of the melt.

Take sample	
Order	MPDV400003
Material	4711-2
Quantity	50.000
Staff badge number	
Cancel Take sample	

Figure: Take sample – C_CHTS

Using this dialog, a sample number is assigned by the HYDRA server and returned as the result to the terminal. A label that includes the sample number is printed. Consequently, the relevant label has to be configured and assigned to the dialog C_CHTS.

The “quantity” field is assigned the quantity of the output buffer by default. This requires, however, that the terminal is connected online with the HYDRA server.

1.5 Cast

Casting means to withdraw a (partial) quantity from the melt.

The 'Cast' dialog box contains the following fields and values:

Order	MPDV400003
Operation	0020
Material	4711-2
Mat. descript.	MPDV Dose
Quantity	50.000
Material buffer	
Batch	PRNZG2Q112
Staff badge number	

At the bottom of the dialog, there are two buttons: 'Cancel' (with a red X icon) and 'Cast' (with a green checkmark icon).

Figure: Casting – C_CHCA

The withdrawn quantity and the target buffer to which this quantity is posted are entered in this dialog. The terminal determines the batch number automatically, provided that the "automatic generation of the batch number" has been enabled in the "MPL" tab of the workplace configuration.

The "quantity" field is assigned the quantity of the output buffer by default. This requires, however, that the terminal is connected online with the HYDRA server.

1.6 Implementation / configuration

The dialogs C_CHPF, C_CHCF, C_CHTS and C_CHCA have to be available at the terminal. The buttons in ctaipbut.ini are configured by using the same IDs:

```
1=C_CHPF,R,perform charging
2=C_CHCF,R,confirm charging
3=C_CHTS,R,take sample
4=C_CHCA,R,cast
```

The section [charge list] is required in the layout configuration of ctaipplay.ini:

```
[charge list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

MAT_VIS=C25,80,L
MATBEZ_VIS=C30,115,L
SOLL_MENGE_VIS=N12.3,78,R,target quantity
EINH=C3,40,L
RES_MENGE_VIS=N12.3,78,R,remaining quantity
VERBR_MENGE_VIS=N12.3,78,R,delivered quantity
MATPUF=C30,80,L
MATPUF_MENGE=N12.3,78,R,remaining quantity
EMAT_MENGE_VIS=N12.3,78,R,input buffer
```

Further configuration details are described in the relevant [procedure](#) for composition functions.